

PROJECT 17: SCHOOL MANAGEMENT SYSTEM

Ngày 30 tháng 3 năm 2026

Project Objective

The objective of this project is to develop a database system to manage students, teachers, classes, subjects, and grades. The system should support daily school operations and provide useful reports for academic and administrative management.

Database and Programming Languages

- **Database Management System:** MySQL
- **Programming Language:** Python

Detailed Project Requirements

System Analysis and Requirements

- Manage student records, teacher assignments, subject offerings, class structures, and academic results.
- Facilitate efficient administration of academic data and school performance tracking.

Main Functionalities

- Student profile management (add/update student info).
- Teacher management and subject specialization tracking.
- Subject and class creation and scheduling.
- Grade entry and academic result reporting.
- Generation of reports on student performance and teacher assignments.

Database Design and Implementation

1. Data Model Design

- Create an ER diagram showing relationships among students, teachers, subjects, classes, and grades.
- Convert the ER model into relational schema including primary and foreign keys.

2. Table Structures

- Students (StudentID, StudentName, BirthDate, ClassID, Address)
- Teachers (TeacherID, TeacherName, Subject, Email)
- Subjects (SubjectID, SubjectName)
- Classes (ClassID, ClassName, TeacherID)
- Grades (GradeID, StudentID, SubjectID, Score)

3. Sample Data

- Add 510 sample rows for each table.
- Include the database schema diagram from MySQL Workbench.

Advanced Database Objects

- **Indexes:** Improve lookup speed for students, teachers, and grades.
- **Views:** Create views for class rosters, top students, and subject-wise performance.
- **Stored Procedures:** Automate student grade updates or class assignments.
- **User Defined Functions:** Calculate GPA or average scores per subject.
- **Triggers:** Auto-update class performance stats when new grades are inserted.

Database Security and Administration

- Create user roles for teachers, academic coordinators, and administrators.
- Secure student and academic records with access control policies.
- Backup strategies for academic data protection and quick recovery.
- Optimize performance for daily use and term-end reporting.

Python Application Development

- **Database Connection:** Use `mysql-connector-python` or `SQLAlchemy`.
- **Student Teacher Modules:** Create functions for data input, grade entry, and class management.
- **Reporting Module:** Generate scorecards, class performance charts, and teacher load summaries.
- **Interface:** Build a CLI or GUI interface for academic staff usage.

Deliverables

- A printed comprehensive report including database design, implementation details, and screenshots. Submit your report into LMS. First pages of your report must attach this PDF file.
- Link to publish Github source code and video presenting on Youtube
- SQL schema, sample data scripts, ER diagrams, and Python code.
- Demonstration of functionalities such as adding income, managing expenses, and generating reports.

Conclusion and Recommendations

- Summarize project outcomes and data integrity checks.
- Recommend future upgrades like attendance tracking, parent access, or mobile integration.

References

- List all academic and technical resources used in the project.